

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Operations & Maintenance Philosophy

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Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPCL	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
CCTV	Closed-Circuit Television
DCS	Distributed Control System
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base

FLM	First Line Maintenance
FME nv	Federal Ministry of Environment
FQI	Flow Quantity Indicator
GSPA	Gas Sales & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
ICSS	Integrated Control and Safeguarding System
IO	Input/Output
IP	Internet Protocol
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NMDPRA	Nigerian Midstream and Downstream Petroleum Regulatory Authority
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License
OSD	Operational Shut Down
PAGA	Public Address and General Alarm
PAS	Process Automation System
PC	Pressure Controller
PCV	Pressure Control Valve
PSV	Pressure Safety Valve
PTW	Permit to Work
RTU	Remote Terminal Unit
SPDC	Shell Petroleum Development Company of Nigeria
VSAT	Very Small Aperture Terminal

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

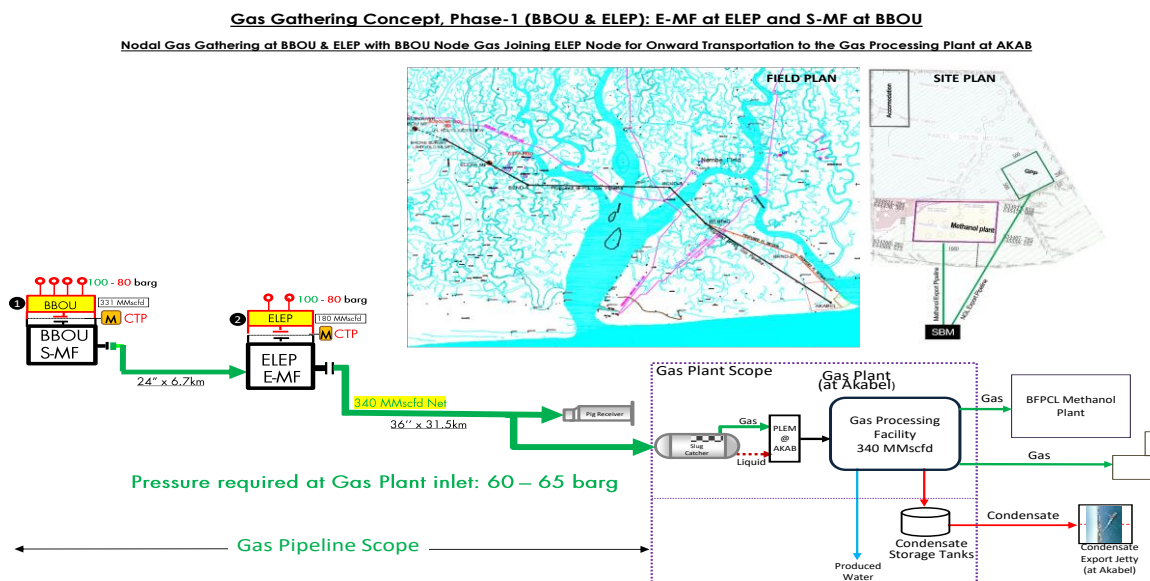


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

To provide a framework for ensuring that the gas gathering pipelines and associated infrastructure design is aligned with the ultimate operational intent and taking into account accessibility, operability and maintainability at optimal cost over the full life cycle.

The FEED shall fully comply with all specified relevant contractual requirements

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Introduction to Operations and Maintenance Philosophy

This Brass Fertilizer Petrochemical Company Limited (BFPCL) Gas Gathering Pipelines and Associated Infrastructure FEED Operations and Maintenance Philosophy defines the guidelines on all necessary operations requirements to be met at every phase of the Asset/project life cycle for the pipelines and the field manifolds. A separate Operations and Maintenance Philosophy will have to be developed for the Methanol Complex but that is outside the scope of this FEED.

The document sets out how the pipelines and manifolds facilities will be operated and managed over its full life cycle to achieve maximum value in accordance with the business objectives.

It will serve as a reference document to the project team and respective design/maintenance contracts in the management of the project with respect to Operability, Maintainability, Reliability, and Monitoring requirements.

This philosophy shall be updated during the detailed engineering phase of this project.

3.1. Key Objectives

- Produce and deliver gas to Custody Transfer points according to agreed quality and plans, safely and without adverse effects on People's health and the environment.
- Ensure high standard of HSSE in all aspects of operations
- Operate the gas gathering pipelines and manifolds in the most cost-effective manner to safeguard People, Environment, Technical Integrity, and Corporate Reputation.

4. Operations Activities

This philosophy is prepared on the premise that there will be a central control room, field logistic base and fit-for-purpose workshop at the Methanol Complex in Akabel to serve the entire field operations including the Gas Gathering Pipelines and Associated Infrastructure.

Operations activities shall be categorized into core and non-core activities which are necessary for efficient management of the pipelines and the manifolds.

The core activities include:

- Operations Management
- Production operation including frontline Maintenance
- Production Measurement, Allocation & Accounting
- Asset Management
- Production Planning
- Operations input to projects

Non-core activities include:

These are support activities, which are either temporary or specialized in nature or those activities that require minimal technical skills.

- Pipeline pigging
- Laboratory services
- Logistics operations
- Heavy lifting using cranes
- Supply of chemicals and other consumables
- Specialized major maintenance etc.

4.1. Process Monitoring and Safeguarding System

There is no Control function for the Pipeline system, only Monitoring and Safeguarding.

Process monitoring and safeguarding system shall be provided for the Pipelines for the purpose of monitoring and protecting the pipelines from adverse operating conditions that could result in harm to people and/or damage to the asset or environment.

Pipeline monitoring and safeguarding systems shall be designed to be fail safe.

Monitoring of systems on each gas manifold will be carried out from the RTU on each gas manifold. The RTU will be interfaced with the DCS on the main plant in Akabel through Fibre Optic Cable with VSAT as backup.

Field instruments should be capable of handling digitally transmitted calibration and maintenance

information (remotely calibrated and tested). The main monitoring system should provide a means to monitor, diagnose problems, and perform surveillance remotely.

- Flow, Temperature and Pressure monitoring system shall be installed on the pipelines for monitoring purposes.
- Transmitters shall be installed to transmit information on the parameters to the Control room.

The field elements and the transmitters shall provide information to the Operators in the Main Control room at Akabel for quick response in order to ensure the pipeline is operating within the design envelope. The Emergency Shutdown System (ESD) and Fire and Gas system (F&G) are combined as an integrated Safeguarding System due to the small number of IOs.

- The Process safeguarding system shall be independent of the process monitoring system.
- The process safeguarding system shall ensure that suitable protection is provided against the maximum pressure (and other operating parameters) that can be generated by the worst credible malfunction.
- Emergency shutdown and blowdown Valves forming part of the process safeguarding system should not have a bypass unless the bypass is locked closed.

4.2. Maintenance Support Function

The maintenance function will be organized as part of the Operations function to perform regular predictive, preventive, and corrective maintenance activities.

Maintenance support shall be provided from BFPCL's facilities at Methanol Complex.

Maintenance support for Field Operation shall include:

- Workshop
- Valves repairs, calibration and servicing
- Inspection
- Provision of maintenance manpower supply

There will be periodic checks of the gas detectors and the frequency will be in accordance with the recommendations of the gas detector vendor.

Check calibration of the instruments shall be done every 6 months.

- The details and types of maintenance support services should be reviewed and agreed between the service providers and service users, mainly the field Operations teams (Production, Electrical, Instrument, Mechanical).
- The maintenance support function will hold manpower call off contract(s) for skilled and semi-skilled personnel for provision of the above Maintenance Support services

4.3. Field Logistics and Transportation

The gas well heads, satellite manifolds and evacuation manifold with operational valves are located in swampy terrain. Regular visits to these manifolds from the Field Logistics Base in Akabel, will be required for operation and maintenance activities. Therefore,

- Operators shall carry out its field operations in a manner that will minimize or eliminate risk to personnel, environment, and assets. Safe and efficient means of transportation shall be provided at all times, in all situations, and under every condition.
- The field logistics philosophy shall cover transportation of personnel and materials into, within, and out of the fields.
- All logistics supports will be provided by BFPCL or through one of its qualified, competent, and experienced service providers.
- All routine material logistics support will preferentially be by marine transportation.
- Movement into, within, and out of fields shall be by marine transport with company provided boats (owned or leased) under armed security escort.

4.4. Health, Safety and Environment (HSE)

- Health, Safety and Environment (HSE) shall be an integral part of BFPCL Gas Gathering Pipelines Operations.
- Operations HSE Cases shall be prepared by the Project team and handed over to Operations as part of Project Hand Over and implemented / owned by Operations.
- All HSE critical roles in Production Operations shall be performed by certified competent staff. A competency assurance process will be developed for implementation to ensure that this is met.
- Contractors providing services for the facility shall develop HSE plans, setting out performance targets and methods to be adopted to achieve set targets.
- Operations activities shall be carried out in compliance with BFPCL HSE policy. The objective will be to assure that the health of employees, contractor personnel and others working in the facility will not be adversely affected by the work.
- A PTW (Permit to Work) system shall be implemented for all non-routine work which includes maintenance and repair, construction and modification, hot work, entry into confined spaces, work in exposed location, pressure testing, heavy lifts, etc.
- The above listing on PTW shall include activities where and when safeguards may be temporarily by passed, or when new hazards may potentially or inadvertently be introduced, e.g., hot work in a producing facility, or in an area where there may be the presence of hydrocarbons, Pipe breaking, etc.
- Selection, approval, and procurement of chemicals will be in accordance with BFPCL approved procedure. Chemicals must be acquired, transported, used and disposed of in

compliance with NMDPRA and FMEEnv regulations.

- No new Chemicals will be accepted from a supplier or vendor without the appropriate Material Safety Data Sheet (MSDS).
- All chemicals will be covered in the Safe Handling of Chemicals (SHOC) cards and enforced through the installation and operation of the facility.
- All personnel will be prevented from exposure to noise in amounts, which may cause noise induced hearing loss, or may affect employee well-being.
- Noise contours and maps shall be developed at the Operate phase of facilities.
- In areas with noise levels above 85-dB (A), the use of hearing protection shall be mandatory.
- Environmental Impact Assessments (EIA) shall be conducted pre and post projects in order to identify baselines and evaluate potential impacts of any project and to minimize negative impact on the existing environmental conditions.
- The facilities shall be designed, operated, and maintained to meet stringent environmental protection objectives within the boundaries of economic, technical, and social constraints.
- The emission and discharge levels from BFPCL operations (PSVs and gaskets) shall be routinely monitored against regulatory limits.
- All fluids disposed into the environment from Saver pits shall be treated to the prescribed quality prior to disposal.
- Flaring and venting of gas (PSVs) shall be eliminated under routine operations or kept to an absolute minimum, within applicable guidelines and regulatory requirements.
- A solid Waste Management System will be put in place prior to commissioning and will document all waste streams, sources, segregation approach, and disposal methods.
- All domestic and industrial wastes generated shall be collected, segregated, and disposed of onshore.
- The Operations Manual for all producing facilities shall contain Emergency Response Procedures covering foreseeable emergency situations (e.g., fire, spill, loss of containment, medical evacuation, security incidents, etc.)
- The Emergency Response Procedure shall be duly signed, regularly reviewed and updated, communicated effectively throughout the organization, including pasting in appropriate places.
- There shall be regular periodic drills involving all cadres of the organization.
- To improve the emergency response capability in the field, formal emergency training and drills are to be periodically carried out. Recommendations and lessons learned shall be built in for improvement.

4.5. Field Security

- Security considerations at all times shall cover personnel, production asset and property, in

this order of priority.

- Security measures must be pre-emptive/proactive and not reactive. Therefore, robust and fit-for-purpose security assessment and plan shall be developed prior to commencement of any field operation and activity and must be implemented accordingly.
- Contractors shall be responsible for arranging and managing their own security in accordance with the rules of engagement as issued by the BFPCL Security organization.
- Evacuation of personnel from a facility or Field Logistic Base (FLB) under security threat must be adequately planned and provided for as part of normal operation. Evacuation plan will be prepared and drills carried out to ensure familiarity with the procedure.
- Armed escort will be provided to accompany and work with teams if they have to travel and work in high security risk locations.
- Field security plan shall be built around local intelligence gathering and use of Government Security Forces (GSF) for enforcement.
- Management shall be responsible for the provision of resources to implement the security plan. Field Management shall be responsible for resource definition and utilization to implement field security.
- Contractors / vendors engaged at field shall submit an acceptable security plan which shall seamlessly tie into the field security and the State security plans.
- In the event of community disturbance and/or illegal tampering, the facilities shall be made safe through automated fail-safe shutdown or remote shutdown initiation from the Main control room by Production Operations personnel.

4.6. Production Chemistry Unit

- A production chemistry unit (including Third Party Service Contractor support) will be required to carry out contaminants analysis, samples and advice on inhibitor injection chemicals and any other chemicals deemed appropriate, as directed by Production Operations in the operating phase.
- Evaluation of effectiveness of various chemical treatments and processes, monitoring, and rating of chemical costs and performances in the operating phase
- BFPCL's laboratory at the Methanol Complex shall handle the analysis required for the gas pipeline.

4.7. Field Communication

Reliable communication network is critical to ensure a safe, reliable and efficient operations and during an emergency situation. An integrated telecommunication network shall be provided for the support of integrated nodal operations and establish data linkages with other associated stakeholders.

All necessary communication infrastructure shall be provided as part of the Gas Gathering

Pipelines (GGPL) Project.

The following strategies will be adopted.

- Field communications will be based on maximum use of reliable commercial communications infrastructure, where available.
- Telecommunication system shall cover all the necessary facilities such as telephones, mobile radio, CCTV, and data highways (ICSS for monitoring and safeguarding purposes).
- LAN, CCTV, ACS to be installed at each gas manifold.
- Access control systems (ACS) and Pipeline intrusion detection system (PIDS) shall be provided for the Manifolds and Pipelines.
- A public address system (PAGA) shall be provided at each manifold to enhance site communication and emergency response preparedness.
- Email/Internet/: Data, IP Cameras/Video Conferencing should be provided. Scope of coverage will include all facilities and locations that have a direct impact on operations objectives.
- Field locations will be linked to the main Communications hub by Fibre Optic Cable (FOC). VSAT system will be used as secondary communications link with a bandwidth range that is sufficient to cover the distance between the gas manifold and the main plant on Brass Island. Communication between the gas manifold and the main plant on Brass Island will depend solely on the project FOC and VSAT systems and will not require third-party telecommunications service providers.
- Fibre optic cables shall be laid together within new pipelines right of way for primary communication link.
- Access to LAN (Local Area network) and WAN (Wide Area Network) shall be available at all field locations via the FOC from the base communication system.
- Approved users will be able to use laptop computers to access emails, websites and other applicable systems as deployed.

5. Operations Readiness and Assurance (OR & A)

This function is to provide operations inputs into project design and installation starting from FEED phase. Also, to ensure operations personnel are adequately trained to operate and maintain the facility. This is to be managed initially by competent Production and Maintenance staff until later in the Operations phase of the gas pipelines.

Other responsibilities include:

- Defines important steps to enable the project to deliver the Operational requirements.
- Hinged on early involvement of the Operations team and preparedness of the People, Systems, Processes, Contracts to enable effective and safe handover of the facilities.
- Outline activities and specific plans required for ensuring all necessary operational input to the respective project phases.
- Ensure that the resources, systems and support infrastructures etc., required to optimally operate and maintain the facilities from commissioning to abandonment are provided at the right levels to meet the overall project objectives.

6. Operations In Project

Production and Maintenance Discipline teams will be involved in all phases of the project development to assist the project team in producing a design that:

- Establishes and meets HSE standard, availability, and reliability requirements at system, sub-system, and equipment levels as identified by studies and in compliance with statutory requirements.
- Minimizes or eliminates the requirements for re-work or carryover of excessive punch-list items into the operational phase. Senior and key Operations staff will be appointed to the project before the start of front-end engineering and design (FEED).
- The main source of Operational support for facilities design shall be operations requirements, operating philosophy definition, project specification reviews, and specific operations input to design activities to ensure safety, operability, and maintainability.
- The production function shall:
- Ensure that production operation technical matters such as plant processes and production field operation practices are adequately and effectively covered into the project, as part of feedback to the organization.
- Provide operations inputs to engineering projects by seconding experienced operations engineers to project teams through the project cycle. This would help ensure that Operations Philosophy, strategies, and requirements are clearly communicated to engineering and project executors in a consistent manner.
- Develop with the design team's assistance, safety, environmental, and regulatory procedures consistent with the Operations Philosophy. Such procedures shall include Fire Plan, Emergency Response Plan, Hydrocarbon Release Plan, Evacuation Plan, Simultaneous Operations Plan (SIMOPS), Safe Operating and Maintenance Procedures, Safe Work Authorization System, Confined Space Entry and Management of Change (MOC) Process. The lowest risk exposure to personnel while working in the field shall be the guiding principle.
- Maintain a well-defined role within the major project execution structure to allow for proper involvement in operability, hazard analysis, HAZOP's, SIL analysis, Alarm rationalization /management database and training of personnel. This role supports the smooth transition and on time delivery of the project.
- Production Operations will staff the project in time to develop the Organization and be fully prepared for the smooth start-up of the facility.
- Develop operations budget to cover expenses for operational readiness from project inception through to startup. This will include the requirements for getting operations and

maintenance personnel to be properly trained and aware of the operating requirements for all equipment and packages.

- Participate in HAZOP Reviews, SIL reviews, Risk Analysis, Alarm rationalization and Human Factor Engineering for non-standard (non-backbone) designs and where new facilities interface with existing systems.
- Participate in Factory Acceptance Tests (FAT's) by sending experienced selected technical personnel to witness or perform test.
- Play a quality assurance role during the project, from inception to hand over ensuring that projects focus on "needs" not "wants" and allow for early training and development of Operations & Maintenance team members and partake in the development of operating and maintenance manuals and procedures.
- Support and provide personnel for commissioning by deciding staffing levels they can provide (project team has lead role in commissioning)
- Responsible for Simultaneous Operations planning and execution
- Assist the Project team in obtaining permits for facilities start-up as may be required
- The Project team shall provide equipment listing including valve types/sizes for the purpose of creating an Asset Register. This information shall be uploaded to the Maintenance Management System for future maintenance planning.
- As part of effort to assure operational readiness, the operational team will contribute, review, and accept:
 - Design Basis Document
 - Scope & Purpose Document
 - Maintenance & Maintainability Document
 - Operations Readiness Plan Document
 - Commissioning Document
 - Start-Up Document

6.1. Commissioning

This section would be aligned with the overall Operations and Maintenance Philosophy of the Methanol Complex.

- Prior to the start of pre-commissioning, the Production and Maintenance team needs to be available and ready. The Permit to Work system will be complied with as appropriate to the project Phase.
- The pre-commissioning activities undertaken after mechanical completion, but prior to commissioning are to prove and validate the functioning of equipment. Pre-commissioning could involve the introduction of fluids into systems but not hydrocarbons.
- Typically, pre-commissioning activities will verify that documentation to support mechanical

completion is in place and not repeat work carried out to achieve mechanical completion. Activities include instrument loop checks, energizing electrical equipment, etc. on the manifold platform equipment.

- Prior to commissioning and introduction of hydrocarbons a Pre-Start-Up audit will verify:
 - Whether facilities are constructed as per design.
 - Operations Philosophy is complied with.
 - Asset Integrity is verified and can be demonstrated.
 - Operations and Maintenance personnel are sufficiently trained and competent.
 - HSE-MS and Operations HSE Case are in place.
 - All required spares and consumables are available (including 2-year operating spares).
 - Maintenance Management System is in place.
- The Pre-Start-Up audit will cover a wider spectrum and will concentrate on the facilities, HSE, systems and operations readiness.
- Upon completion of a satisfactory Pre-Start-Up audit and readiness for start-up, operatorship of the plant will be transferred from the Commissioning team to Operations. At this point the Project team shall retain the responsibility for the integrity and performance of the facility up to the final acceptance.
- The Project will supply all relevant procedures, materials and consumables, and supporting manpower, including technical support, to allow safe operation of the facilities to final acceptance.
- A commissioning strategy, high level logic and start-up sequence (Precedence Network) shall be developed at an early stage in the detailed design by the Project team.
- Besides the normal plant start-up procedures for the facilities for both planned and unplanned shutdowns the Project team through the CONTRACTOR shall develop the one-off commissioning start-up procedure.
- Start-up procedures will be reviewed and approved by BFPCL before start-up.
- The commissioning team, supplemented by BFPCL operations, will operate the non-hydrocarbon systems during commissioning activities – Operations shall only take the responsibility for introduction of hydrocarbons, and operating the associated process systems. Coordination between operations and the commissioning team is crucial.
- Upon completion of a satisfactory Final Acceptance Audit and closure of all critical items, the ownership of the Asset can be transferred to the Operator, which would then close the Execution phase of the project.
- Operations staff shall join the pre-commissioning team as part of “hands-on” training on site.
- Operations shall be responsible for the introduction of hydrocarbons under the direction of the commissioning engineer.
- Factory Acceptance Test (FAT) shall be carried out in the vendor's yard prior to shipment of

any equipment.

- Commissioning tests will cover comprehensive process and mechanical conditions that are required during production. A detailed test agenda, special test equipment required, and vendor participation and responsibilities shall be defined in the commissioning plan document ahead of time.
- The commissioning plan will cover system/loop/equipment items testing using standard check sheets and will define, in collaboration with the vendors, consumables and spares needed during commissioning, the commissioning process with milestones, and documentation requirements.

6.2. Facilities Handover

The following criteria (to be aligned with overall project hand-over requirements) shall form the basis for hand-over and formal acceptance of facilities by Operations from the commissioning phase:

- A successful 72-hour sustained performance test consisting of a continuous and stable operation on automatic operation and monitoring of the pipelines and equipment without technical interruption. This closely monitored period shall be followed by a 3-month reliability test to ensure that all parts of the facility operate as per design specifications.
- All data incorporated for maintenance routines, Operations HSE case, procedures, operating / maintenance manuals, and routines necessary to support operations shall be in place.
- A commissioning hand-over checklist (to be used in the hand-over process) detailing outstanding items that may exist at hand-over phase from commissioning. They shall be identified as punch list items classified into categories 'A' and 'B' as follows:
- Category 'A' punch list items (Interim Handover) are items that must be repaired, completed or corrected before the system can be handed over or accepted.
- Category 'B' punch list items (Final Handover) will be items that can be repaired, completed, or corrected at a later date and allow the system to be operated without jeopardizing safety of people, asset, or the environment.
- The initial operating spares and special maintenance tools shall be available on site prior to hand-over. An audit shall be conducted on the status of all spare parts prior to handover from project to operations.

6.3. Handover Documentation

This section would be aligned with the overall Handover Documentation of the Methanol Complex.

- 3 hard copies and an electronic copy of a Pipelines Operating and Maintenance procedures manual shall be submitted for the gas pipelines.

- All documents shall be provided in an electronic format compatible for direct input into the Electronic Documentation Management System (EDMS) platform.
- Hand over documentation to the operations and maintenance department will include:
 - HAZOP Report and recommendations.
 - Completed commissioning check sheets during commissioning.
 - Vendor documentation consisting of general operating and maintenance instructions, spare parts, drawings/lists, equipment data sheets, equipment operating and performance test criteria.
 - Operating and start-up manuals.
 - Production Operating Procedure Manuals (POPM).
 - Completed Computerized Maintenance Management System, data sheets, and Maintenance Job Routines (MJR's).
 - Maintenance Manual.
 - Maintenance Reference Manual.
 - As-built drawings.
 - Electrical single line drawing.
 - Facility HSE Case.
 - Safe Handling of Chemicals (SHOC) manual.
 - Pipeline Operating and Emergency Repair manual.

To ensure smooth transition and hand over of the documents to the Operations / Maintenance departments, it is foreseen that different revisions of the above documents may be handed over at different phases of the project to suit operational needs.

7. Management Of Change In Project & Operations Phases

No change shall take place to the process, facility, procedure, or people without adequate consideration of the impact of such change to HSE, operations, and the system.

- All changes shall be duly approved by an appropriate approval body after due consideration.
- All change requests shall be submitted to the approval body through a secretary to the body, after consideration, review, and support of end users, technical authority, and a sponsor.

8. Operating Strategy

8.1. Field Manifolds

The two gas gathering manifolds for phase 1 of this project are the Bubouwe Bou (BBOU) and Elepa manifolds. The BBOU manifold and Elepa manifold will not be manned but will be attended for operations and maintenance purposes.

Access shall be provided to the manifolds through the water ways from the gas plant to allow personnel travel from the plant location to the manifolds for operational reasons by boat.

The manifold shall have elevated platforms with perimeter fencing to prevent unauthorized entry to the platforms.

To ensure the security of the valves and equipment on the platforms, a surveillance team shall be engaged from the community, to be supported by the government security agents. Appropriate monitoring technology shall be considered to provide advance warning of intruders.

Communication system shall be provided between the platforms and the plant control room to facilitate passage of information to/from the locations.

Gear assisted valves shall be designed / installed on the manifolds for easy operation. Pig launcher and receiver shall be installed on the manifolds as part of pipeline design for regular pipeline cleaning.

Primary Power supply for the manifolds shall be through the fibre optic cable from the Akabel power generation system. Consideration will be given to other back up power system e,g, diesel generator or the Solar system, to ensure higher reliability.

Shelter shall be provided on the platform to protect personnel from weather elements.

8.2. Staff Resourcing

The workforce shall be made up of skilled, competent, and well-motivated personnel to achieve excellence in operation and maintenance.

To establish the manning levels, the following among other factors shall be considered:

- Manning philosophy (core and non-core activities)
- Outsourcing philosophy
- Minimum visit practices
- Minimum intervention practices
- Staff requirements and experience levels
- Daily operations and maintenance
- Relief cover (e.g., leave, training, medical, etc.)

- Shift cycle

8.3. Training Requirements

- Operations staff that will be required to operate and maintain BFPCL gas gathering pipelines and the manifolds will be provided with both technical and emergency response training to existing standards so that they will be competent to safely and efficiently operate the facilities.
- Training will be provided for staff during the project execution phase.
- Identified operations staff shall be nominated and made available, full time, during the pre-commissioning and commissioning phases of this gas pipelines project. In addition, appropriate personnel shall be made available on a part-time basis during the detailed design phase of the project.
- The training will begin during Detailed Engineering Design and will be completed at the pre-commissioning stage so that each staff is adequately prepared for the commissioning phase.

8.4. Field Logistics Base

- Full-fledged accommodation will be provided in Akabel near the Methanol Complex for operations and maintenance personnel that will operate the gas gathering pipelines as part of the BFPCL facilities for quick response to operational demands.
- Permanent accommodation facility will be designed to provide a comfortable and pleasant living environment away from home for the staff.

8.5. Mode of Operations

- The mode of operations for all regular field crews shall be determined during the Detailed Design phase.
- Multi- skilled (Production, Mechanical, Electrical & Instrumentation) teams shall operate the facilities and carry out first line maintenance work, supported by discipline specialists for major overhaul via healthcare contract.
- Accommodation will be provided for routine and non-routine personnel who have to work in the facility and if the work requires them to stay overnight at the location due to operational requirements.

9. Hydrocarbon Measurement & Accounting

- All incoming and outgoing hydrocarbon streams to and from field facilities shall be measured, using appropriate technology and practice to meet contractual, production, fiscal, custody transfer, royalty, and regulatory requirements as per the measurement design philosophy.
- Appropriate hydrocarbon accounting procedures shall be established to cater for present and all future evacuation processes.

9.1. Metering System

- The overall production measurement philosophy shall be developed and agreed as part of the Gas Sales Agreement.
- Gas measurements for fiscal / custody transfer, allocation, and production measurements shall be in accordance with the respective NUPRC/NMDPRA guidelines for gas and liquid measurements.
- The NUPRC/NMDPRA preferred / approved meter type for fiscal / custody transfer measurements for liquid are Coriolis meters and others while for gas is Daniel Orifice.
- Other meter types (V-cone, Ultrasonic, etc.) may also be considered, however, an approved waiver shall be sought and obtained from NUPRC/NMDPRA.
- All fiscal and allocation metering stations shall be designed as a dual stream facility to ensure maximum availability. Thus, one metering stream is always available to enable continued gas export metering during routine maintenance, inspection and validation of any of the metering facilities.

9.2. Incoming Gas Stream

Allocation metering shall be provided at BFPCL plant inlet manifold for gas measurement of all incoming pipelines.

9.3. Condensate Export

Custody transfer metering shall be provided at delivery points of condensate export pipelines with a meter at injection points.